Project Milestone 2: Plan and Build Your App with Streamlit

CODE REPOSITORY LINK: https://github.com/isaactabor21/ds5023-dashboard

Purpose: In this milestone, you move from concept to construction. You will decide the main pages of your application, sketch each page before coding, specify the visual elements on every page, and then build those pages as a working Streamlit multipage app. Your design decisions must be grounded in the visual design principles (hierarchy, emphasis, contrast, alignment, whitespace, Gestalt, typography, color, scale & proportion, balance, consistency, unity & variety) and core design frameworks (Norman, Shneiderman, Tognazzini, Rams) covered in class. You will also implement data visualizations and conduct a design self-review using the Visual Design Review Checklist.

Important: Throughout this milestone, reference your Milestone 1 (M1) persona, design principles, usability targets, and heuristic evaluation findings. Your design decisions should be traceable back to M1. If there are changes from M1, you don't need to update your M1 deliverable, but document what changed and why in your Design Justification.

Deliverables (submit as one PDF + code repository link):

1. **Page Architecture Plan (with sketches):** Sketches for all pages (minimum 4) showing layout, key UI elements, and how users move between pages. Include a simple list naming each page and its purpose.
2. **Working Streamlit Multipage Application:** Multipage application with functional navigation, input widgets, and display elements. Code must run without errors.
3. **Data Visualizations:** At least 2 charts/visualizations integrated into your app. Charts should respond to user input or filters.
4. **Design Justification:** document why you made your design decisions and what changed from sketches to implementation.
5. **Visual Design Review:** Review your own app against your milestone 1 design principles and usability targets. What's working? What needs improvement? (1 page)

1. Page Architecture Plan (20 pts)

Before writing any code, sketch all pages/screens of your application. You can use paper, a whiteboard, a tablet, or any simple tool (such as PowerPoint, Google Slides, or draw.io). The goal is rapid iteration, not polished design. Sketching first helps you plan the layout, navigation, and user flow before getting distracted by implementation details.

- **Page List**

Your app must have at least 4 distinct pages. You decide what pages make sense for your interactive experience.. For each page, provide:

- Purpose (what users accomplish on this page)
- User goal it supports (connect to your M1 persona)

DS 5023: Page Architecture Plan

- Page List (4 distinct pages):

Page 1: Home & Search

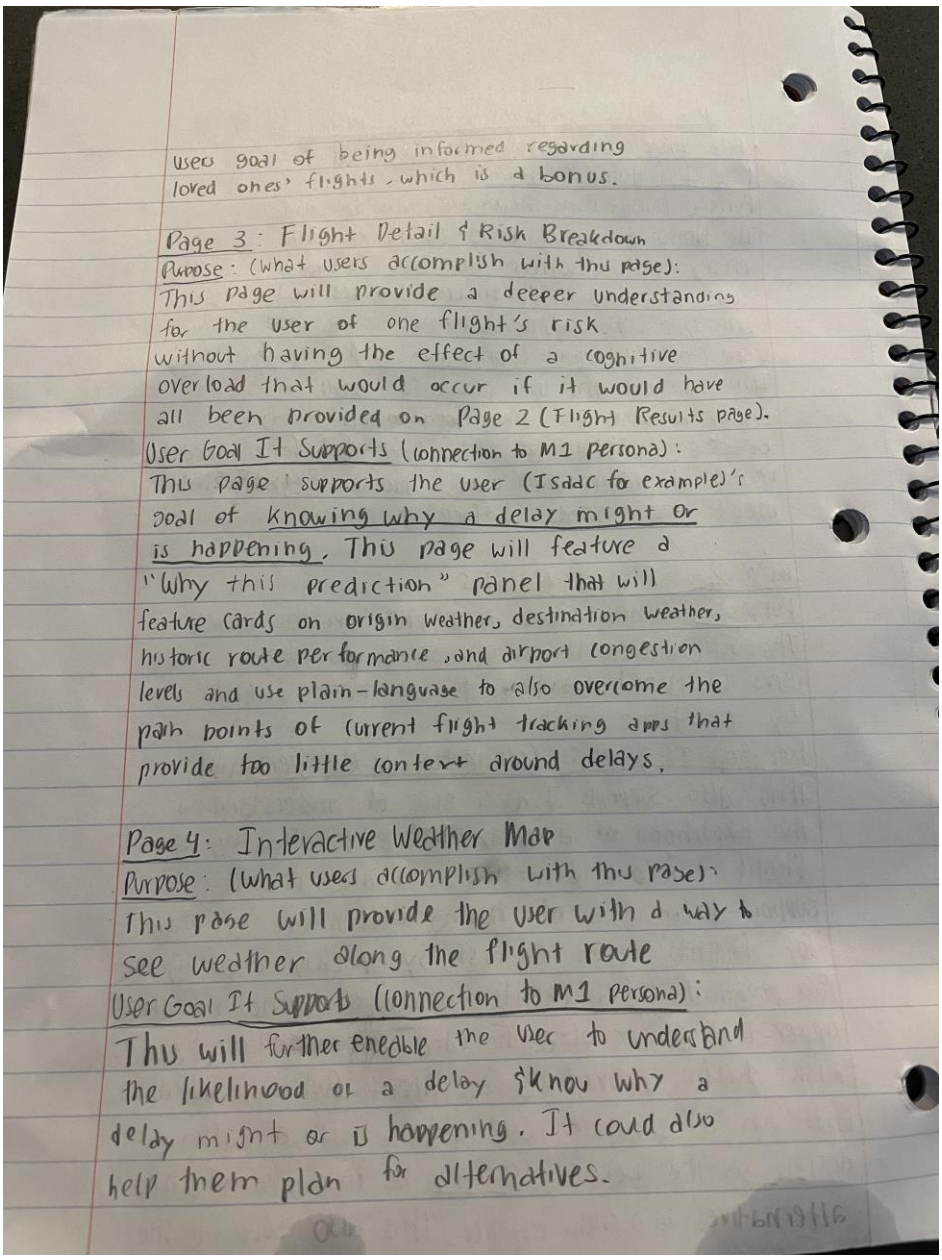
Purpose (what users accomplish on this page):
The Home & Search page serves as the entry point where the user (Isaac) sets his origin, destination, dates and trip type (One-way, Round, Multi).

User Goal It Supports (connection to M1 Persona):
This supports the user (Isaac)'s goal of understanding the likelihood of a delay before purchasing a flight & before departure because it allows the user to browse flights that meet their needs before purchasing.

Page 2: Flight Results

Purpose (what users accomplish on this page):
The Flight Results page serves to allow users (Isaac for example) to compare flights by on-time risk.

User Goal It Supports (connection to M1 persona):
This also supports Isaac's goal of understanding the likelihood of a delay before purchasing a flight & before departure, but also it will support their goal of having alternative options for flights based on risk, by quickly showing % probabilities of flights being on time in the upper-left corner of each card, color-coded by risk & also providing an alert (when applicable) with an alternative suggestion with a calm design, so the user can quickly be informed of risks & alternatives in a calm manner. This also supports the



users goal of being informed regarding loved ones' flights, which is a bonus.

Page 3: Flight Detail & Risk Breakdown
Purpose: (what users accomplish with this page):
This page will provide a deeper understanding for the user of one flight's risk without having the effect of a cognitive overload that would occur if it would have all been provided on Page 2 (Flight Results page).
User Goal It Supports (connection to M1 persona):
This page supports the user (Isaac for example)'s goal of knowing why a delay might or is happening. This page will feature a "Why this prediction" panel that will feature cards on origin weather, destination weather, historic route performance, and airport congestion levels and use plain-language to also overcome the pain points of current flight tracking apps that provide too little context around delays.

Page 4: Interactive Weather Map
Purpose: (what users accomplish with this page):
This page will provide the user with a way to see weather along the flight route
User Goal It Supports (connection to M1 persona):
This will further enable the user to understand the likelihood of a delay & know why a delay might or is happening. It could also help them plan for alternatives.

- **Sketches**

For each page, create a sketch that shows:

- Layout structure — where elements are positioned on the page and approximate sizing/proportions.
- Key UI elements — buttons, inputs, sliders, visuals, navigation
- Navigation — how users move between pages

Start with sketching multiple versions quickly. Your first idea is rarely your best.

- **Design Annotations**

Annotate each sketch with at least one visual design principle that guided a placement or grouping decision.

Examples: "Proximity: grouped filter controls together in sidebar so they read as one unit" or "Hierarchy: metric cards are largest at top because they're the first thing the user needs to see."

Example of Screens/Pages for Common App Types

App Category	Typical Screens
E-Commerce App	Home, Product Listing, Product Detail, Cart, Checkout, Order Confirmation, Order History, Profile, Auth, Spending Stats
Analytics Dashboard	Dashboard Overview, Analytics/Reports, Data Tables, Detail View, Charts, Settings, Admin Panel, Auth
Task / Project Management	Dashboard, Data Table, Detail View, Calendar, Activity Feed, Progress/Reports, Settings, Auth
Booking / Reservation App	Home, Search/Results, Detail View, Calendar, Checkout/Payment, Trip Stats, Confirmation, Order History, Auth
Content / Media App	Home/Feed, Explore, Media Player, Content Detail, Profile,
	Search, Listening/Viewing Stats,Notifications, Auth
Educational / Learning App	Home, Course Catalog, Progress Dashboard, Lesson View, Progress Dashboard, Quiz/Assessment, Profile, Auth
Health / Fitness App	Dashboard, Activity Log, Calendar, Detail View, Settings, Profile, Progress/Trends, Onboarding, Auth

Air Aware

Home & Search

Proximity tells the user trip-type buttons are grouped

- One-way
- Round-Trip
- Multi-City

From:
DCA – Washington, D.C.

To:
MSP – Minneapolis-Saint Paul

Depart
Dec 16

Passengers
1 Adult

Emphasis. Scale hierarchy: put bigger & at top so user sees this first & as most important on page.

Search Flights ← Error prevention
(disable until complete / if clicked without entering flight display message "Please enter all flight search details")

↑ goes to Flight Results page when clicked

Recent searches
DCA → LGA Feb 27
DCA → MIA Mar 4

← Efficiency (Tognazzini): optimized for experienced users to use quickly.

Home	Flight Results	Flight Risk Breakdown	Weather Map

error prevention in navigation tabs
(display "Please search for a flight to access this page" if hover or click)

Air Aware

Flight Results

← Back DCA → MSP - Dec 16

Filters: Risk ▾ Time ▾ Date ▾

Horizontal Bar: On-Time Prob. by Flight

UA 441 [ ] 91%

AA 802 [ ] 32%

Pie Chart: Risk of Delay Distribution Today

- Low (67-100%)
- Medium (33-66%)
- High (0-33%)

Emphasis & Hierarchy through color (67-100% will be green, 33-66% will be orange, 0-32% will be red). This will pull users eye to prob. flight is on-time because this is most important to their goal.

91% On-Time Probability

9:15AM - 12:20PM - 3h05m

United 441

MSP → DCA

nonstop

hierarchy through position: top-left position ensures user sees % first.

32% On-Time Probability

2:30PM - 5:28PM - 2h58m

American 888

MSP → DCA

nonstop

9 more flights ↓

Scale & Emphasis: Warning will be in red & big text to signal importance

⚠ MSP Weather alert: Winter storm Dec 16

Better option found.

Dec 15 DCA → MSP - 94% On-Time

Color Contrast: Alert will be red, Better option in green

Proximity: Alert & Better option will appear close & with consistent cards to show they're related.

Home | Flight Results | Flight Risk Breakdown | Weather Map

Error Prevention: If on dashboard message "Click on a probability button to see its risk breakdown".

Air Aware

Flight Detail & Risk Breakdown

Dec 16 American 888 DCA → MSP 2:30 PM – 5:28 PM – 2h 58m

32 %

On-time probability · HIGH RISK

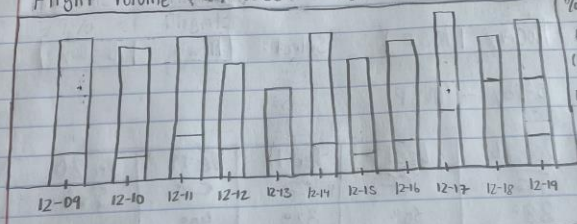
Why this prediction?

Scale & emphasis: largest letters b/c this is the focal point & user goal for this page.

Weather at DCA
Clear skies, 34°F. Low wind. No icing risk.

⚠ **Weather at MSP**
Heavy snow, 28°F. High wind. Icing risk

Flight Volume & Delay Risk Stats DCA → MSP (Dec 9–Dec 16)

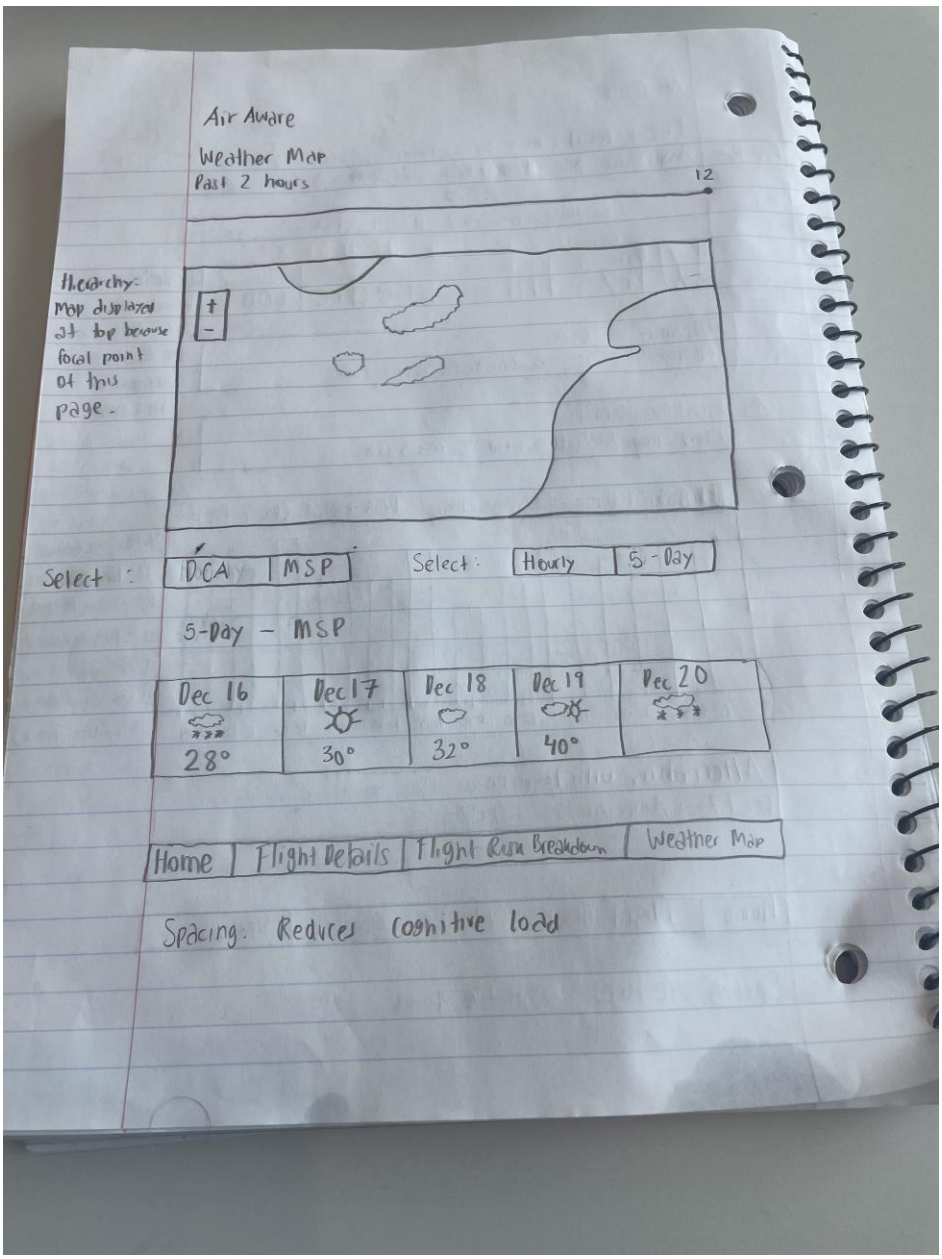


%= On-Time Prob.
□ = High Risk (0-33%) or Delayed
□ = Medium Risk (33-67)
□ Low Risk (68= 100%/ On-Time)

Alternatives with lower risk
Dec 17 · American 888 - 94%

Home	Flight Results	Flight Risk Breakdown	Weather Map

Spacing: reduces cognitive load



2. Basic Application Implementation (40 pts)

Build all planned pages as a working Streamlit multipage application. Your implementation should reflect the page architecture, sketches, and visual design specifications from the sections above.

- **Layout & Structure (15 pts)**
- Use at least 2 layout primitives: st.sidebar, st.columns or st.expander
- Organize controls logically (e.g., filters in sidebar, results in main area)
- Include clear page title and section headers

- **Input Widgets (15 pts)**
- Include at least 5 distinct interactive widgets across your pages (e.g., st.selectbox, st.slider, st.text_input, st.date_input, st.button, st.multiselect, st.toggle, st.radio). Widgets must meaningfully affect what the user sees , not just exist on the page.
- Widgets should be functional and affect displayed output
- Use appropriate widget types for the data being collected
- **Performance and Display Elements (10 pts)**
- Use st.cache_data where appropriate. Briefly note (in code comments or your writeup) where and why you applied caching.
- Display data using st.dataframe or st.table
- Include appropriate feedback messages (st.success, st.warning, st.info, st.error)
  - Use st.metric for key summary statistics where appropriate

Code must run without errors. Test your application thoroughly before submission.

3. Data Visualization (15 pts)

Implement interactive data visualizations that serve your persona's goals. Visualizations must be integrated into your Streamlit pages, not standalone scripts.

- Include at least 2 different chart types (e.g., line chart, bar chart, pie chart, scatter plot)
- Charts should be interactive or respond to user filter selections
- Visualizations must support your persona's goals
- Use Streamlit's built-in charts or integrate Plotly for advanced features **Chart Selection Guide**

Chart Type	Best For
Line Chart	Trends over time (e.g., spending by month, progress over weeks)
Bar Chart	Comparing categories (e.g., expenses by category, tasks by status)
Pie/Donut Chart	Part-to-whole relationships (e.g., budget breakdown)
Scatter Plot	Relationships between two variables (e.g., time spent vs. score)
Heatmap	Patterns across two dimensions (e.g., activity by day and hour)

4. Design Justification (10 pts)

Write a brief document explaining your design decisions:

- How your sketches evolved from your Milestone 1 (persona, design principles, and heuristic evaluation findings)
- Why did you choose specific widgets for each page?
- How does your layout and navigation support the user's primary tasks ?
- Why did you select your chart types and how they help your persona ?
- What changed (if any) from your original sketches during implementation and why?
- What trade-offs (if any) did you make (e.g., simplicity vs. features, aesthetics vs. functionality)?

Our initial idea from Milestone 1 was aiming to show everything in the same tab, but that is a little bit of information overload, and it likely would overwhelm a user with too much going on. Throughout this milestone we evolved to show everything, but more spaced out. Instead of everything being displayed at once, there's a clear process for a user to explore a flight, the costs and risks involved with a flight, and how they can proceed with that option, while additionally being given alternatives. The display is set up so that a user can find a flight they want, explore the potential options, identify the risks associated with that flight, potential alternatives, and a radar map to see what the weather will be like.

Our home page is simple and understandable. A user can select the type of flight they are looking to book (one-way, round-trip, multi-city), the airports they are hoping to depart from and travel to, and their travel dates. They additionally have their recent searches at their convenience. Our flight results tab makes it easy for a user to filter criteria such as risk status, flight timing, and cost. This allows a user to select the flight(s) that fit their important criteria, while making them aware of the risks associated with the flight they are taking. The risk analysis and weather radar tabs simply show the viewer in depth why the risks are present, and overall, what's going on with their flight. Our weather radar allows a user to see a weather map for the area.

Our layout allows a user to follow through on the tasks best suited for their needs. They can easily and efficiently select the airports they travel to and from, when they travel, identify the time, costs, and risks associated with the potential flights. The layout and visuals present will allow a user to look at how flights are expected to perform, for example, if a user sees that there is only one flight that's low risk from the pie chart, they will be able to focus their attention on that flight and it makes their process much easier to accomplish their goals. The historical bar chart for flights shows how reliable a given flight has been historically and how effective our model has performed in terms of risk analysis. This allows our user to trust our product and its reliability. The weather radar additionally offers them the opportunity to look at what the weather might be at the time of their flight.

We were able to implement our design based on our sketches, but there are some trade-offs with our design. We need to work on integrating our models with real-time flight data, according to the specific airports, locations, flights, and timing involved. Additionally, we need to validate the weather radar map to ensure that it's able to navigate a user to the weather forecast for that location and flight. It's a challenging inclusion. Lastly, a user might not use the full app to make their decision. For example, having the risks listed in depth as well as historical accuracy in the risk analysis tab might be missed because a user will make their decision based on the flight

results tab. Something to update in the future could be to update the webpage to display the following tab after selecting a flight, rather than ensuring the user clicks for the risk analysis.

Our current implementation relies on a limited dataset containing specific airport routes (e.g., DCA → MSP). As a result, the search functionality does not yet validate whether a user's selected airports exist within the underlying database. While users can input different airports, the application may still return default flight results because matching logic has not been fully implemented.

In future milestones, we will improve usability and system transparency by constraining user input through validated airport selections. For example, the search interface could use dropdown menus populated only with airports contained in our dataset, or display a message indicating when no matching routes are available. This aligns with Norman's principles of constraints and visibility and Shneiderman's guideline to prevent errors and provide informative feedback, ensuring users clearly understand when a search is successful versus unsupported by the data. We currently are also limited because access to real-time commercial flight data APIs is restricted and typically requires paid enterprise-level subscriptions, so in future milestones we may research how to overcome this because right now we are limited to historical data that we plan to try to integrate in future milestones.

5. Visual Design Review (15 pts)

Conduct a self-review of your implemented pages using the Visual Design Review Checklist covered in class. This is your quality check: does the built app match your design intent?

Review your app at four levels:

- **Detail View (3 pts):** Check typography (correct fonts, sizes, weights), spacing (consistent padding and margins), colors (match your palette), and icon/element alignment. Report at least 2 specific observations (things done well or issues found and fixed).
- **Page-Level View (4 pts):** Check visual hierarchy (does the eye flow to the most important element first?), alignment (do elements snap to a consistent grid/column structure?), whitespace (is there breathing room, or is the page cluttered?), and content fit (no overflow, truncation, or orphaned elements). Report at least 2 specific observations.
- **Function-Level View (4 pts):** Check that key states are handled: loading states, error states, and success states. Check form validation and that users always know where they are in the app. Report at least 2 specific observations.
- **App-Level View (4 pts):** Check that patterns are consistent across all pages (same button styles, same card formats, same color usage). Check that your navigation is coherent and the overall information architecture is logical. Report at least 2 specific observations.

For each level, describe what you found and what you fixed (if any). The purpose is to demonstrate that you can critically evaluate your own work using the principles you've learned.

Our app is not ready for full deployment yet. There are a lot of potential changes to make it easier and more efficient for an active or new user to use the app, as well as functional errors that need to be modified or fixed.

From a detail view perspective, the font and background currently set in place fails the **contrast** principle. We need to either change the background or font to ensure that a user can easily discern between the text and understand what they are being presented with. The most important aspect of our app (risk levels) are consistently displayed along the same color scale based on the risk level across the app.

At the page-level there are whitespace and hierarchical issues present. There is a lot of white space present along the flight results tab that can distract the viewer from the really important things to account for in the visual. The flight risk, details, and cost are all present but are of varying font sizes and improving upon the visual hierarchy could help make it easier for a viewer to identify the flights they like. Potentially to use a F-pattern or Z-pattern design for flight information.

Right now the multi-city and round-trip functional logic aren't working yet. There is only the option to select a departing flight, and a user isn't able to identify the risks and costs associated with a return flight or additional flight. There isn't the option to select multiple airports for a multi-city trip yet. In terms of functionality on our app, it's extremely valuable and helpful that our flight results and risk analysis pages offer the user feedback to understand what they need to do in order to utilize the app.

From the app-level, Shneiderman's principles are important to include for our app. Particularly, his principles that users can receive informative feedback, easily reverse actions, and keep users in control. Our app effectively offers feedback based upon usage of the flight results and risk analysis pages, prompting the user that they need to search a flight in order to look at the details. To add to the app, it would be helpful to direct users to the tabs they are expected to use based on their use of the app – to improve usability. For example, upon searching for a flight, a user will be taken to the flight results page rather than needing to search that on their own, as well as being taken to the risk analysis page upon finding a desired flight. Lastly, it could additionally be of value to the user for them to easily reverse actions – for example to use the back arrow to return to a previous tab, or be able to search for prior flights they've looked at (which our app allows in the home page).

Reference: Visual Design Principles

Use this reference when making design decisions and writing your justification. You do not need to submit this separately, but your work should demonstrate these concepts.

Visual Design Principles

Principle	Application
Hierarchy	Size, color, and placement indicate importance
Emphasis	Draw attention to key elements through contrast
Contrast	Differentiate elements through color, size, or weight
Alignment	Create visual connections through consistent positioning
Whitespace	Give elements room to breathe; reduce cognitive load
Consistency	Same patterns for same functions throughout
Typography	Font choices and size hierarchy create structure

Color	Semantic meaning (success/warning/error) and brand identity

Gestalt Principles

Principle	How to Apply
Proximity	Place related items close together
Similarity	Use same colors/shapes for similar functions
Common Region	Use containers or backgrounds to group elements
Continuity	Align elements to create visual flow
Closure	Users perceive complete shapes from partial elements
Figure-Ground	Make important elements stand out from background

Core Design Frameworks

Framework	Principle
Norman	Visibility, feedback, constraints, mapping, consistency, affordance
Shneiderman	Strive for consistency, seek universal usability , offer informative feedback, design dialogs to yield closure, prevent errors and offer recovery, permit easy reversal of actions, keep users in control, reduce short-term memory load
Tognazzini	Autonomy, discoverability, efficiency, explorable, Fitts's Law, latency, learnability, track state
Rams	Good design is innovative, useful, aesthetic, understandable, unobtrusive, honest, long-lasting, thorough down to the last detail, environmentally friendly, and involves as little design as possible